Table of Contents

Exp. No.	DATE	Title	Page No.	STAFF SIGNATURE
1		EXPLORATION OF THE HADOOP INSTALLATION		
2		HADOOP IMPLEMENTATION OF FILE MANAGEMENT TASKS		
3		IMPLEMENT OF MATRIX MULTIPLICATION WITH HADOOP MAP REDUCE		
4		RUN A BASIC WORD COUNT MAP REDUCE PROGRAM TO UNDERSTAND MAP REDUCE PARADIGM		
5		IMPLEMENTATION OF K-MEANS CLUSTERING USING MAP REDUCE		
6		INSTALLATION OF HIVE ALONG WITH PRACTICE EXAMPLES		

EX: 1 **DATE:-**

EXPLORATION OF THE HADOOP INSTALLATION

AIM:

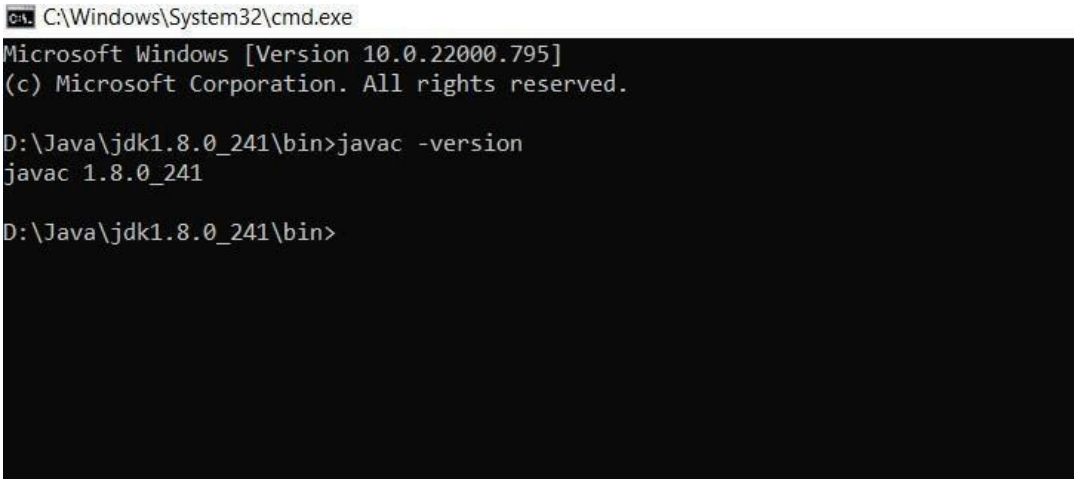
To create an Exploration of the hadoop environment.

ALGORITHM:

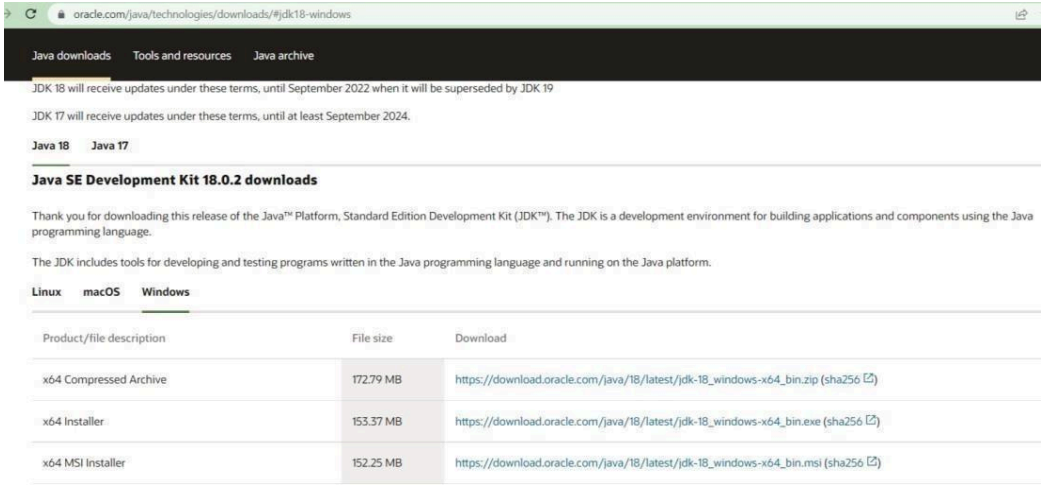
STEP1: To install Hadoop, First you should have Java version 1.8 in your system.

STEP 2: Check your java version in this command on command prompt

javac –version

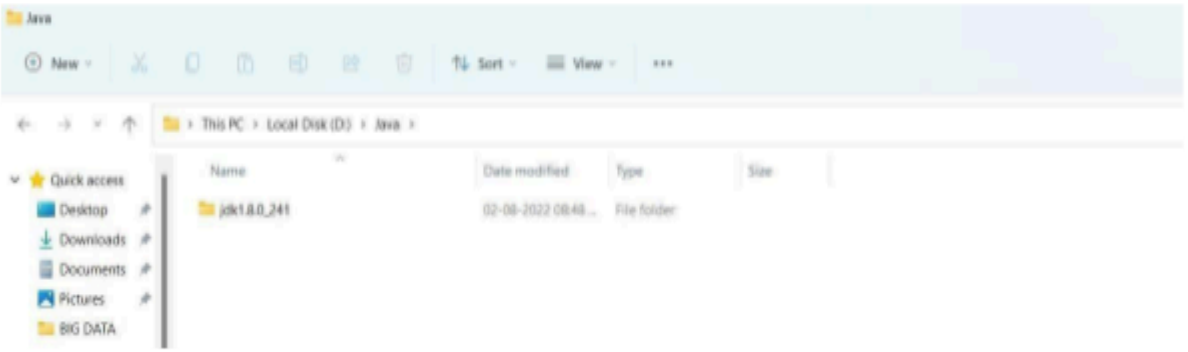


STEP3: If java is not installed in your system



STEP4: Download the file according to your operating system. Keep the java folderdirectly under the local disk directory

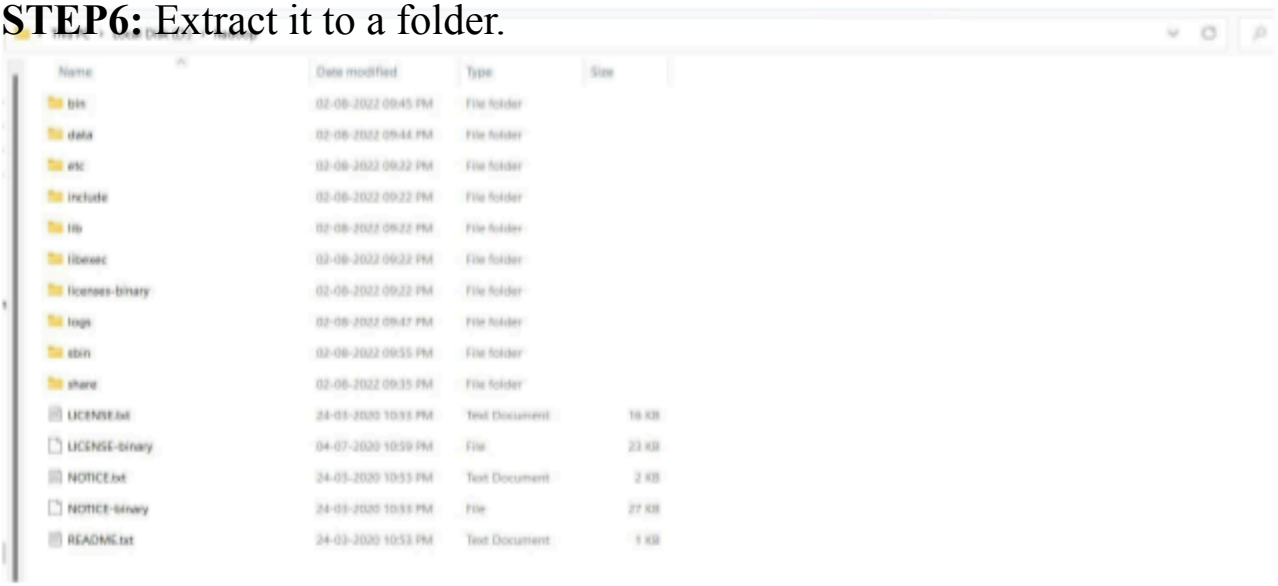
(D:\Java\jdk1.8.0_241\bin)



STEP5: After downloading java version 1.8, download hadoop version 3.1

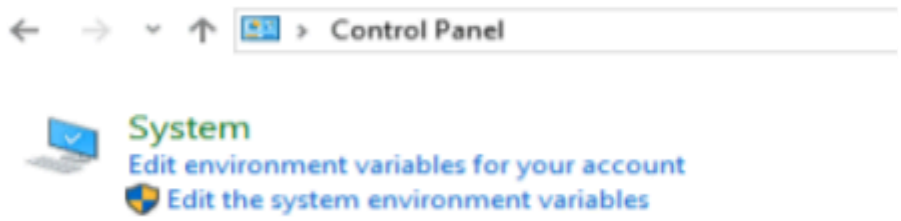
Hadoop Link:
https://archive.apache.org/dist/hadoop/common/hadoop-3.3.0/hadoop-3.3.0.tar.gz

STEP6: Extract it to a folder.

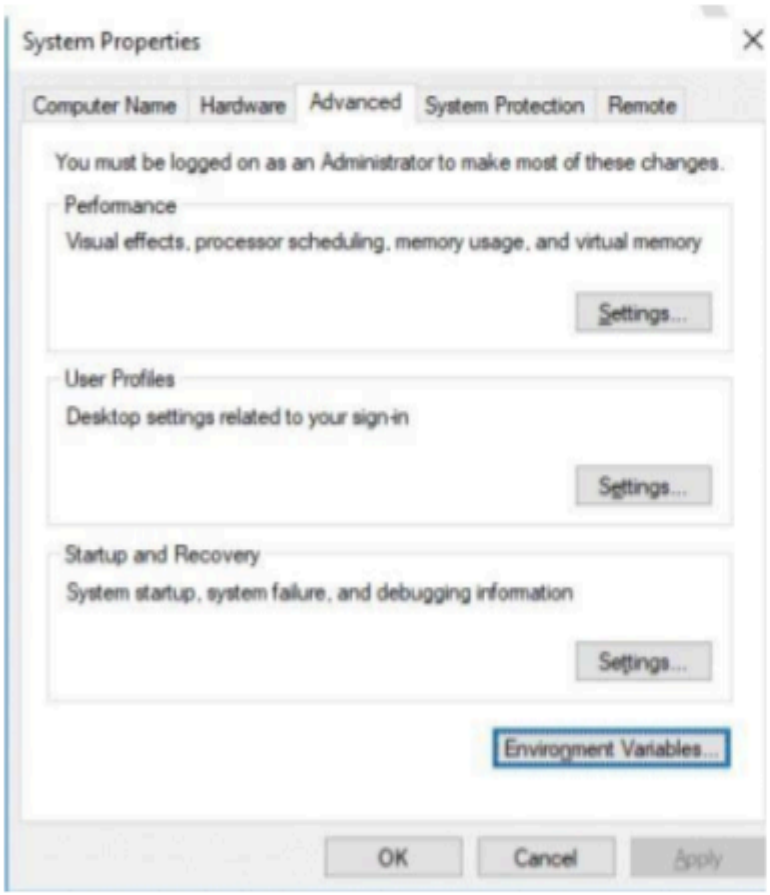


SETUP SYSTEM ENVIRONMENT VARIABLES

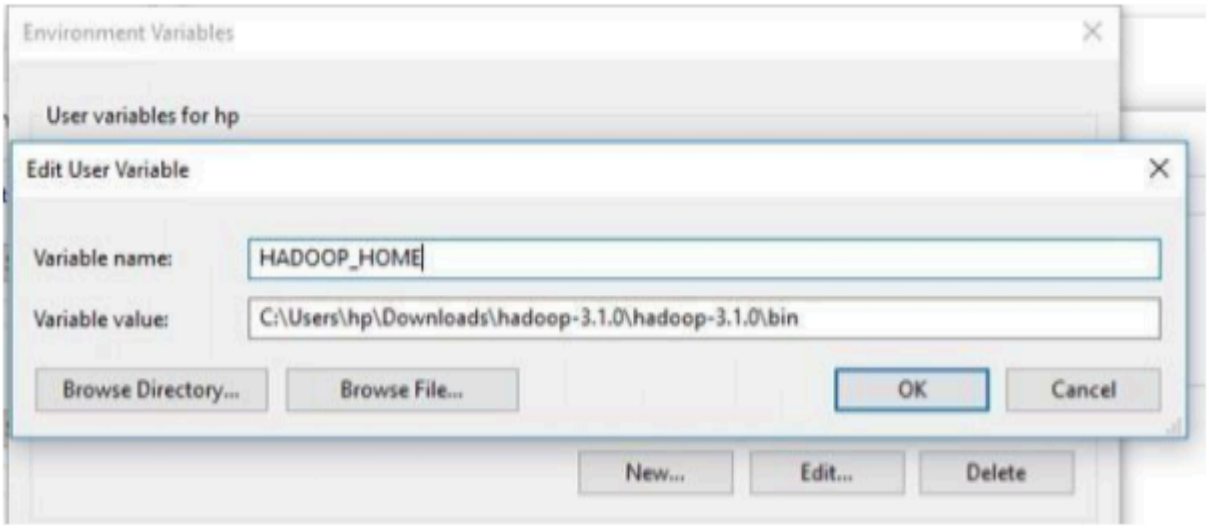
STEP7: Open control panel to edit the system environment variable



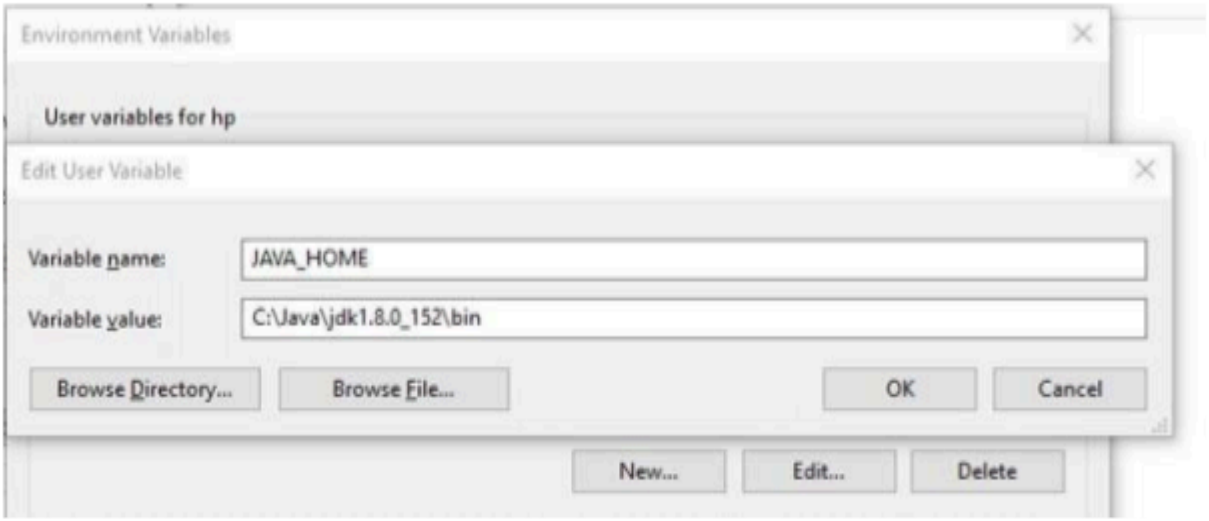
STEP8: Go to environment variable in system properties



STEP9: Create a new user variable. Put the Variable name as HADOOP_HOME and Variable value as the path of the bin folder where you extracted hadoop

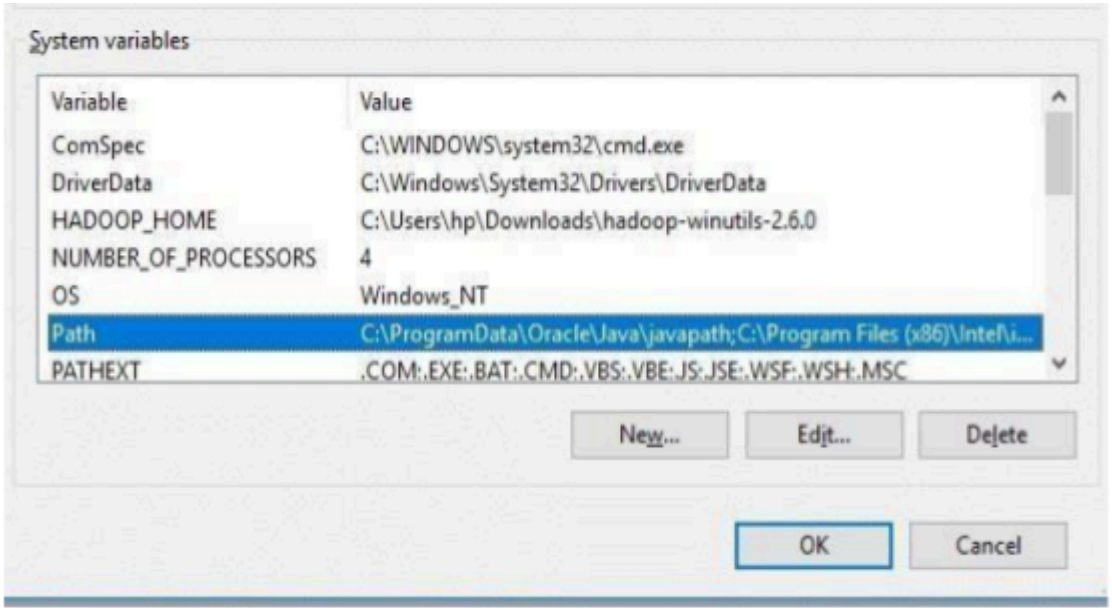


STEP10: create a new user variable with variable name as JAVA_HOME and variable value as the path of the bin folder in the Java directory.

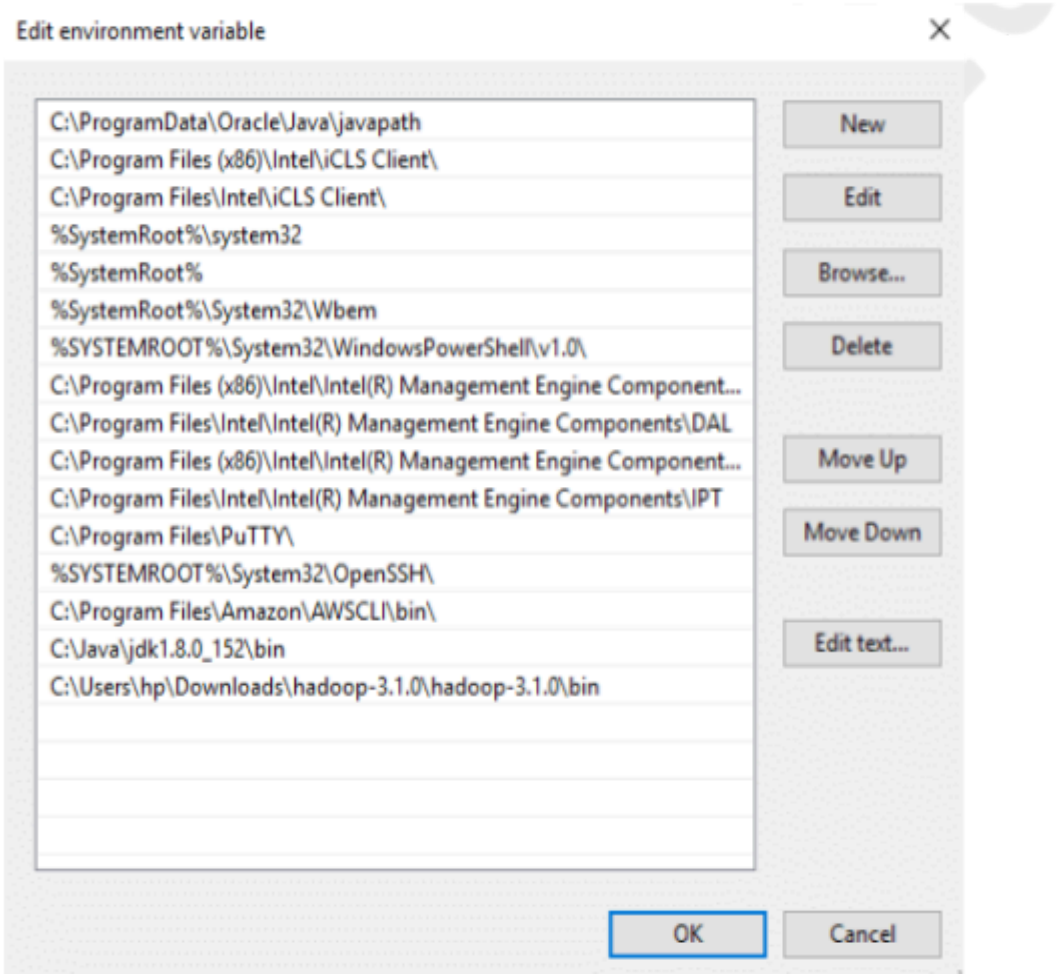


STEP11: Now we need to set Hadoop bin directory and Java bin directory path in system variable path.

STEP12: Edit Path in system variable



STEP13: Click on new and add the bin directory path of Hadoop and Java in it.



RESULT:

Thus the Program has been successfully completed.

EXP:2 **DATA:**

Hadoop Implementation of file management tasks,

such as Adding files and directories, Retrieving files and Deleting files

1.

Create a directory in HDFS at given path(s).

SYNTAX :-

hadoop fs -mkdir <paths>

Example:

hadoop fs -mkdir /user/saurzcode/dir1 /user/saurzcode/dir2

2.

List the contents of a directory.

SYNTAX :-

hadoop fs -ls <args>

Example:-

hadoop fs -ls /user/saurzcode

3.

Upload and download a file in HDFS.

SYNTAX :-

hadoop fs -put:

Copy single src file, or multiple src files from local file system to the Hadoop data file system

SYNTAX:-

hadoop fs -put <localsrc> ... <HDFS_dest_Path>

Example:

hadoop fs -put /home/saurzcode/Samplefile.txt /user/

saurzcode/dir3/**Download:hadoop fs -get:**

Copies/Downloads files to the local file system

SYNTAX:-

hadoop fs -get <hdfs_src> <localdst>

Example:

hadoop fs -get /user/saurzcode/dir3/Samplefile.txt /home/

4.

See contents of a file

Same as unix cat command:

SYNTAX:-

hadoop fs -cat <path[filename]>

Example:

hadoop fs -cat /user/saurzcode/dir1/abc.txt

5.

Copy a file from source to destination

This command allows multiple sources as well in which case the destination must be a directory.

SYNTAX:-

hadoop fs -cp <source> <dest>

Example:

hadoop fs -cp /user/saurzcode/dir1/abc.txt /user/saurzcode/ dir2

6.

Copy a file from/To Local file system to HDFS CopyFromLocal

SYNTAX:-

hadoop fs -copyFromLocal <localsrc> URI

Example:

hadoop fs -copyFromLocal /home/saurzcode/abc.txt /user/ saurzcode/abc.txt

Similar to put command, except that the source is restricted to a local file reference.

copyToLocal

SYNTAX:-

hadoop fs -copyToLocal [-ignorecrc] [-crc] URI <localdst>

Similar to get command, except that the destination is restricted to a local file reference.

7.

Move file from source to destination.

Note:- Moving files across filesystem is not permitted.

SYNTAX:-

hadoop fs -mv <src> <dest>

Example:

hadoop fs -mv /user/saurzcode/dir1/abc.txt /user/saurzcode/ dir2

8.

Remove a file or directory in HDFS.

Remove files specified as argument. Deletes directory only when it is empty

SYNTAX:-

hadoop fs -rm <arg>

Example:

hadoop fs -rm /user/saurzcode/dir1/abc.txt **Recursive version of delete.**

SYNTAX:-

hadoop fs -rmr <arg>

Example:

hadoop fs -rmr /user/saurzcode/

9.

Display last few lines of a file.Similar to tail command in Unix.

SYNTAX:-

hadoop fs -tail <path[filename]>

Example:

hadoop fs -tail /user/saurzcode/dir1/abc.txt

10.

Display the aggregate length of a file.

SYNTAX:-

hadoop fs-du <path>

Example:

hadoop fs -du /user/saurzcode/dir1/abc.txt.

RESULT :-

Thus the Program Implementation of file management tasks has been successfully completed